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Theories for Deep Learning

Advanced Topics in Deep Learning

Raghavendra Selvan

Tenure-Track Assistant Professor
Dept. of Computer Science (ML Section)
Pioneer Centre for Artificial Intelligence, Denmark
ELLIS Member

raghav@di.ku.dk

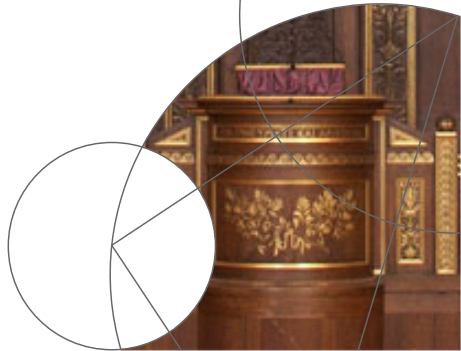
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<https://raghavian.github.io>

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<https://saintslab.github.io>

September 1, 2026



Learning is a tug of war

Every learning method **minimises** a sum of *two opposing terms*:

$$\text{objective}(h) = \underbrace{\text{loss of } h \text{ on } S \text{ (e.g. } \widehat{R}_S(h))}_{\text{fit term} \quad \downarrow \text{ as capacity } \uparrow} + \underbrace{\text{cost to specify } h \text{ (e.g. } \|\mathbf{w}\|, \text{ code length)}}_{\text{complexity term} \quad \uparrow \text{ as capacity } \uparrow}$$

where training data $S = \{(x_i, y_i)\}_{i=1}^n$; a candidate model $h \in \mathcal{H}$; training error $\widehat{R}_S(h) = \frac{1}{n} \sum_i \ell(h(x_i), y_i)$.

- Drive the training data **fit** term to 0 alone: memorise the training set, generalise poorly.
- Minimise the model **complexity** term alone: the constant predictor, generalise poorly.
- The claim of years of statistical learning theory: there is a *sweet spot* in between.

Topic-1 of ATDL

Can deep learning still be explained in this view? If not, are there other plausible explanations?



Most learning methods can be seen as pulling between **fit** and **complexity** terms

Framework	Fitness term	Complexity term
Regularised ERM	$\widehat{R}_S(h)$	$\lambda \Omega(h) \quad (\ \mathbf{w}\ _2^2, \ \mathbf{w}\ _1, \dots)$
Generalisation bounds	$\widehat{R}_S(h)$	$\sqrt{\text{capacity}(\mathcal{H})/n}$
MDL / two-part code	$L(S \mid h)$ (residual bits)	$L(h)$ (model bits)
Bayesian evidence	$-\ln p(S \mid \hat{\mathbf{w}})$	$-\ln \frac{p(\hat{\mathbf{w}})}{\sqrt{\det H}}$ (Occam factor)
PAC-Bayes	$\widehat{R}_S(Q)$	$\text{KL}(Q \parallel P)/n$

ERM: empirical risk minimisation · MDL: minimum description length · PAC: probably approximately correct.

Same framing everywhere: **fit** + **complexity**. A decreasing + an increasing term $\Rightarrow$ the best model sits at *finite* capacity, not at either extreme.

Rissanen, (1978), "Modeling by shortest data description"

Hinton et al., (1993), "Keeping the neural networks simple by minimizing the description length of the weights"

McAllester, (1999), "Some PAC-Bayesian Theorems"

For the Bayesian evidence, $-\ln p(S \mid \hat{\mathbf{w}})$ is the training negative log-likelihood (NLL; cross-entropy or mean-squared error) at the fitted weights $\hat{\mathbf{w}}$; $-\ln \frac{p(\hat{\mathbf{w}})}{\sqrt{\det H}}$ is

— $\ln$ of the Occam factor which prefers solutions in flat regions ($\det H$ small).

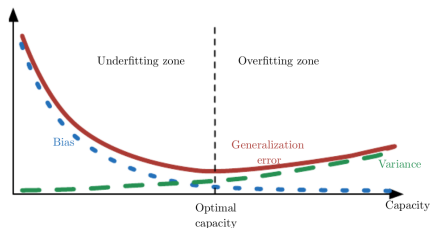


The Classical Verdict: Do not fit to training data perfectly

- **Bias–variance:** $\mathbb{E}_S[(y - \hat{h})^2] = \sigma^2 + \text{Bias}^2 + \text{Var}$, with $\text{Var} \uparrow$ in capacity.
- **Structural risk minimisation** (Vapnik): pick capacity k minimising $\hat{R}_S + \text{penalty}(k)$.
- **MDL** (Rissanen): pick h minimising $L(h) + L(S | h)$.
- All land at a *finite* capacity, where

$$\hat{R}_S(\hat{h}) > 0$$

One must *underfit* the training data a little to generalise. Zero training error is past the sweet spot.



classical bias–variance trade-off

Vapnik, (1998), *Statistical Learning Theory*

Rissanen, (1978), “*Modeling by shortest data description*”

Grünwald, (2007), *The Minimum Description Length Principle*



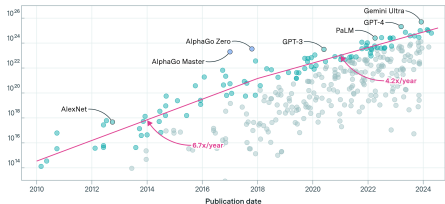
Deep learning has changed regime

Training compute of frontier models

EPOCH AI

Training compute (FLOP)

Outliers Non-frontier Frontier 96 frontier models



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epoch.ai

Training data of notable LLMs

Training dataset size (tokens)

199 models



EPOCH AI | CC-BY

epoch.ai

Figures: Epoch AI (CC-BY). (Sevilla et al. 2022) ; (Villalobos et al. 2022) .

- Frontier compute $\sim 4\text{--}5\times/\text{year}$, data $\sim 3.7\times/\text{year}$, parameters likewise; > 10 orders of magnitude since AlexNet.
- Neural Networks are trained to **zero training error / loss**, often in ≈ 1 pass over the data, with $p \gg n$.

Sevilla et al., 2022, "Compute trends across three eras of machine learning"

Villalobos et al., 2022, *Trends in Training Dataset Sizes*



Interpolation is common with deep learning models

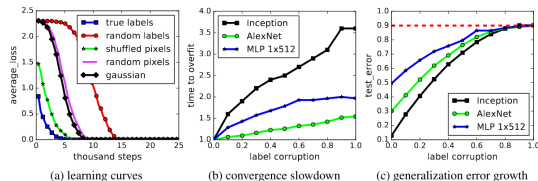
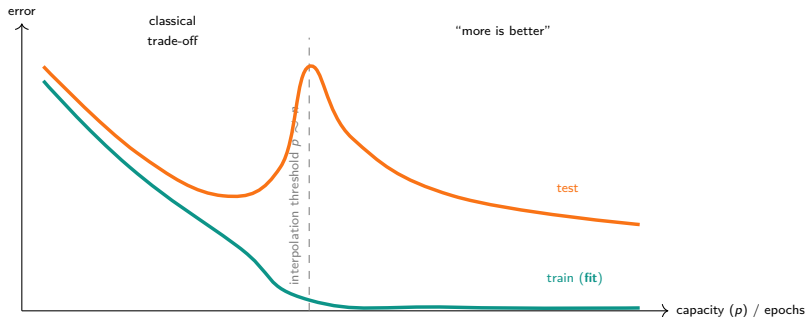


Figure 1: Fitting random labels and random pixels on CIFAR10. (a) shows the training loss of various experiment settings decaying with the training steps. (b) shows the relative convergence time with different label corruption ratio. (c) shows the test error (also the generalization error since training error is 0) under different label corruptions.

- Modern nets drive the **fit** term to **zero**: $\widehat{R}_S(\hat{h}) = 0$, every example fit exactly; and they still generalise.
- The *same* architecture fits **randomly permuted labels** to zero error (Zhang et al.): the class-**complexity** term (its Vapnik–Chervonenkis dimension d_{VC} , its Rademacher complexity $\mathfrak{R}_n$) is pinned at its maximum.
- So a penalty that only sees $\mathcal{H}$ cannot tell the generalising solution from the memorising one.



Double descent: the trade-off curve has a second half



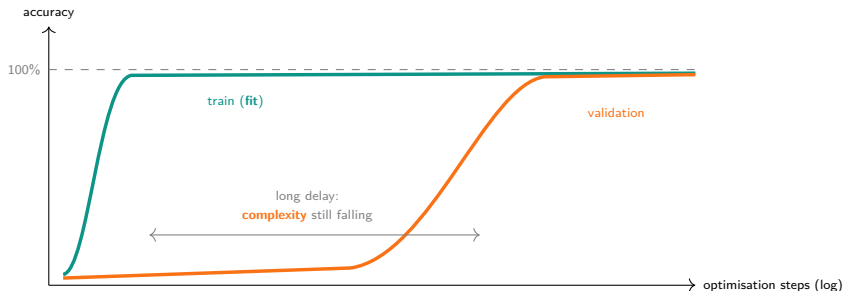
- The bias–variance U is only the **left half**. Past interpolation, test error *descends again*.
- The quantity governing the second descent is *not* p or $d_{VC}(\mathcal{H})$: it is a property of the specific interpolant (e.g. its norm: $\hat{\beta} = X^+y$).

Belkin et al., (2019), “Reconciling modern machine-learning practice and the classical bias–variance trade-off”

Nakkiran et al., (2021), “Deep double descent: Where bigger models and more data hurt”



Grokking: generalisation after the fit term is done



- **fit** term hits 0 in a few hundred steps. Generalisation appears 10^2 – 10^4 steps later.
- In between, a *second* quantity keeps decreasing (weight norm; description length of the representation). Generalisation correlates with these factors.



... and more of the same flavour

- **Benign overfitting:** $\widehat{R}_S(\hat{h}) = 0$ on noisy data, yet $R(\hat{h}) \rightarrow R^*$ (Bartlett et al. 2020) .
- **Implicit bias:** Gradient descent on separable data $\rightarrow$ the *max-margin* solution, due to the optimiser, not the loss, picks h (Soudry et al. 2018) .
- **Scaling laws:** $R \approx R_\infty + (N_0/N)^{\alpha_N} + (D_0/D)^{\alpha_D}$ (Kaplan et al. 2020)
- **Emergence:** abrupt capability jumps with scale (Wei et al. 2022)

In every case: *effective complexity* of the trained $\hat{h} \neq$ *nominal complexity* (p, d_{VC}) of $\mathcal{H}$.

Bartlett et al., 2020, "Benign overfitting in linear regression"

Soudry et al., 2018, "The implicit bias of gradient descent on separable data"

Kaplan et al., 2020, "Scaling laws for neural language models"



Tug of war, no more.

Generalisation in deep learning

- At interpolation the **fit** term is close to 0: it no longer chooses between solutions.
- Among the infinitely many $\{h : \widehat{R}_S(h) = 0\}$, some generalise and some do not.
- So generalisation is decided *entirely* by the **complexity** term, and it must be a property of the **solution the algorithm selected**, not of $\mathcal{H}$.

Q1 Which complexity functional of $\hat{h}$? (norm? information? description length? flatness?)

Q2 Does stochastic gradient descent (SGD) *implicitly minimise* it (or is the optimiser the regulariser)?

Q3 Why is the classical variance penalty simply *absent* once $\widehat{R}_S = 0$?



Six attempts to model the **fit** and **complexity** terms for deep learning

OPENING THE BLACK BOX OF DEEP NEURAL NETWORKS VIA INFORMATION

Opening the black box of Deep Neural Networks via Information

Ravid Schwartz-Ziv
Edmond and Lily Safra Center for Brain Sciences
The Hebrew University of Jerusalem
Jerusalem, 91904, Israel

RAVID.ZIV@MAIL.HUJI.AC.IL

Naftali Tishby*
School of Engineering and Computer Science
and Edmond and Lily Safra Center for Brain Sciences
The Hebrew University of Jerusalem
Jerusalem, 91904, Israel

TISHBY@CS.HUJI.AC.IL

Editor: ICR1-CI

Abstract

Despite their great success, there is still no comprehensive theoretical understanding of learning with Deep Neural Networks (DNNs) or their inner organization. Previous work (Tishby and Ziv

PAPER: ML 2021

Deep double descent: where bigger models and more data hurt*

Preetum Nakkiran^{1,3,*}, Gal Kaplun^{1,4}, Yamin Bansal^{1,4},
Tristan Yang¹, Boaz Barak¹ and Ilya Sutskever²

¹ Harvard University, United States of America² OpenAI, United States of America³ Email: preetum@cs.harvard.edu

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Abstract. We show that a variety of modern deep learning tasks exhibit a ‘double-descent’ phenomenon where, as we increase model size, performance first gets worse and then gets better. Moreover, we show that double descent occurs not just as a function of model size, but also as a function of the number of training epochs. We study the above phenomena by defining a new complexity measure we call the *effective model complexity* and conjecture a generalized double descent with respect to this measure. Furthermore, our notion of model complexity allows us to identify certain regimes where increasing (or even overfitting) the number



Computing Nonvacuous Generalization Bounds for Deep (Stochastic) Neural Networks with Many More Parameters than Training Data

Gintare Karolina Džugaitė
Department of Engineering
University of Cambridge

Daniel M. Roy
Department of Statistical Sciences
University of Toronto

Abstract

One of the defining properties of deep learning is that models are chosen to have many more parameters than available training data. In light of this capacity for overfitting, it is remarkable that simple algorithms like SGD reliably obtain solutions with low test error. One roadblock to explaining these phenomena in terms of implicit regularization, structural properties of the solution, and/or features of the data is that many learning bounds are qualitatively vacuous when applied to networks learned by SGD in this ‘‘deep learning’’ regime. Logically, in order to explain generalization, we need nonvacuous bounds. We obtain an idea by Lanford and Canzani (2001), who

for trained neural networks in the modern deep learning regime where the number of network parameters exceeds the number of training examples.

The bounds we compute are data dependent, incorporating millions of components optimized numerically to identify a large region in weight space with low average empirical error around the solution obtained by stochastic gradient descent (SGD). The data dependence is essential: indeed, the VC dimension of neural networks is typically bounded below by the number of parameters, and so one needs as many training data as parameters before (vacuous) PAC bounds are nonvacuous, i.e., before the generalization error falls below 1. To put this in concrete terms, on MNIST, having over 72 hidden units in a fully connected first layer yields vacuous PAC bounds. Evidently, we are operating far from the worst case ob-

Towards Understanding Grokking:
An Effective Theory of Representation Learning

Zining Liu, Omid Kisseam, Niklas Nohle, Eric J. Michaud, Max Tegmark, Mike Williams
Department of Physics, Institute for AI and Fundamental Interactions, MIT
ezliu@mit.edu, mkiss@mit.edu, nnohle@mit.edu, mtegrark@mit.edu, mwilliams@mit.edu

Abstract

We aim to understand grokking, a phenomenon where models generalize long after overfitting their training set. We present both an analytic and a microscopic analysis of phase diagrams describing learning performance across hyperparameters. We find that generalization originates from structured representations whose training dynamics and dependence on training set size can be predicted by our effective theory in a top setting. We observe empirically the presence of four learning phases: memorization, grokking, nonmonotonic, and confusion. We find representation learning to occur only in a ‘‘Goldilocks zone’’ (including memorization and grokking) between memorization and confusion. We find on transformers the grokking phase steps closer to the memorization phase

Published as a conference paper at ICLR 2018

A BAYESIAN PERSPECTIVE ON GENERALIZATION AND STOCHASTIC GRADIENT DESCENT

Samuel L. Smith* & Quoc V. Le
Google Brain
sml@l3bu.google.com

ABSTRACT

We consider two questions at the heart of machine learning: how can we predict if a minimum will generalize to the test set, and why does stochastic gradient descent find minima that generalize well? Our work responds to (Zhang et al. 2016), who showed deep neural networks can easily memorize randomly selected training data, despite generalizing well on real labels of the same inputs. We show that the same phenomenon occurs in small linear models. These observations are explained by the Bayesian evidence, which penalizes sharp minima but is invariant to model parameterization. We also demonstrate that, when one holds the learning rate fixed, there is an optimum batch size which maximizes the test set accuracy. We propose that the noise introduced by small mini-batches drives the parameters towards minima where evidence is large. Interpreting stochastic gradient descent as a stochastic differential equation, we identify the ‘‘noise scale’’ $\sigma = \sqrt{\frac{1}{N} - \frac{1}{B}}$ as $\propto \sqrt{B/N}$, where N is the learning rate, N the training set size and B the batch size. Consequently the optimum batch size is proportional to both the learning rate and the size of the training set, $B_{opt} \propto N$. We verify these predictions empirically.

1 INTRODUCTION

This paper shows Bayesian principles can explain many recent observations in the deep learning

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Binarized neural networks converge toward algorithmic simplicity: empirical support for the learning-as-compression hypothesis

Eduardo Y. Sakabe^{1,2}, Felipe S. Abrahão^{1,2,3,4}, Alexandre Simões⁵,
Esther Colombini¹, Paula Costa¹, Ricardo Gudwin¹ and
Hector Zenil^{1,2,3,4,5,6,7,8}

¹Faculdade de Engenharia Elétrica e de Computação (FEEL), Universidade Estadual de Campinas, Campinas, São Paulo, Brazil; ²Algoritmos Dinâmicos Lab, King's College London, London, United Kingdom; ³Centro de Engenharia, Computação e Modelagem de Sistemas (CECM), Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; ⁴Unidade de Engenharia de Sistemas, Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; ⁵Unidade de Engenharia de Sistemas, Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; ⁶Unidade de Engenharia de Sistemas, Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; ⁷Unidade de Engenharia de Sistemas, Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; ⁸Unidade de Engenharia de Sistemas, Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil

Understanding and controlling the complexity of neural networks is a central challenge in machine learning, with implications for generalization, optimization, and model capacity. While most approaches rely on entropy-based loss functions



Paper 1 — Schwartz-Ziv & Tishby: information theory

Opening the black box of deep neural networks via information

- Use mutual information between input-and-representation, and representation-and-output.
- Both forces are *information* quantities of a hidden layer T : **fit** $= I(T; Y)$ (bits kept about the label), **complexity** $= I(X; T)$ (bits kept about the input).
- Claim: training runs it in two stages: a short fit phase ($I(T; Y) \uparrow$), then a long *compression* phase ($I(X; T) \downarrow$).



Paper 2 — Dziugaite & Roy: PAC–Bayes

Computing nonvacuous generalization bounds for deep (stochastic) networks with many more parameters than training data

- Predict with a noisy network $h \sim Q$. **fit** $= \hat{R}_S(Q)$; **complexity** $= \text{KL}(Q \| P)$: bits to describe the trained weights given a prior P fixed before data.
- *Optimise the sum directly* ($R(Q) \leq \hat{R}_S(Q) + \sqrt{\frac{\text{KL}(Q \| P) + \ln(2\sqrt{n}/\delta)}{2n}}$) and get a real number: a guarantee < 1 for a net with $p \gg n$.



Paper 3 — Smith & Le: Bayesian inference & SGD dynamics

A Bayesian perspective on generalization and stochastic gradient descent

- **fit** = $-\ln p(S \mid \hat{\mathbf{w}})$; **complexity** = the Occam factor = $-\ln \frac{p(\hat{\mathbf{w}})}{\sqrt{\det H}}$, which favours *flat* minima.
- The *optimiser* does the tug of war: mini-batch noise acts like temperature, $g \approx \epsilon N/B$ (learning rate $\times N$ / batch), pushing the iterate into the broad, high-evidence minima.



Paper 4 — Nakkiran et al.: empirical phenomenology

Deep double descent: where bigger models and more data hurt

- The classical **fit-complexity** curve (the bias–variance U) is only half: test error goes down, *up*, then down again as capacity grows.
- **Effective model complexity (EMC)** $= \max\{n : \hat{R}_S \leq \epsilon\}$ is the largest sample the procedure can still fit; the peak sits at $\text{EMC} \approx n$, not where classical theory puts the “sweet spot”.



Paper 5 — Liu et al.: Explaining grokking

Towards understanding grokking: an effective theory of representation learning

- Grokking: the **fit** term hits 0 fast, but generalisation appears much later while a *second* quantity, the structure of the learned representation, keeps improving.
- An effective theory (a few order parameters) predicts *when* the structured (short-description) representation forms.



Paper 6 — Sakabe et al.: algorithmic information theory

Binarized neural networks converge toward algorithmic simplicity

- Zooms in on the **complexity** side alone: an estimate of the algorithmic (Kolmogorov) complexity $\mathbb{K}(\mathbf{w})$ of the binarised weights, via the Block Decomposition Method (BDM) — the *model-bits* term of MDL, not a two-part code and not the **fit** residual.
- Controls **fit** by looking only at the pre-overfitting regime (validation-loss plateau). Finding: $\mathbb{K}(\mathbf{w})$ *decreases* there — SGD drifts toward algorithmically simple weights — and BDM tracks the loss better than Shannon entropy.



Six candidates for the **complexity** term

#	Paper	proposed complexity term for the trained network
1	Shwartz-Ziv & Tishby '17	$I(X; T)$ — information the representation keeps about the input
2	Dziugaite & Roy '17	$\text{KL}(Q\ P)$ — bits of the solution beyond the prior
3	Smith & Le '18	$\frac{1}{2} \ln \det H$ — sharpness / (inverse) Bayesian evidence
4	Nakkiran et al. '21	effective model complexity vs. n — measured, not derived
5	Liu et al. '22	(dis)order of the learned representation (effective theory)
6	Sakabe et al. '26	$\mathbb{K}(\mathbf{w})$ — algorithmic complexity of the weights

All are **description-length-flavoured** (Blier et al. 2018) . None is proven to be *the* one; and none covers every phenomenon.



Open questions

- Is there *one* right complexity functional, or is it problem-dependent?
- Is it *always* a compression / description length? (KL, $I(X; T)$, $\mathbb{K}$, MDL.)
- Is SGD the regulariser? Does the optimiser implicitly minimise complexity?
- Why is the classical variance penalty *absent* at interpolation?
- Can *one* framework predict double descent *and* grokking *and* scaling laws — or do we need a new “theory” per phenomenon?



Exercise session

- 01_when_classical_theory_breaks: memorising random labels; model-wise double descent.
- 02_grokking: grokking on modular addition;

Zhang et al., (2017), *"Understanding deep learning requires rethinking generalization"*

Nakkiran et al., (2021), *"Deep double descent: Where bigger models and more data hurt"*

Liu et al., (2022), *"Towards Understanding Grokking: An Effective Theory of Representation Learning"*



Take-aways

- Statistical learning theory spent decades on the **fit** – **complexity** trade-off and concluded: *do not fit perfectly*.
- Deep learning fits perfectly and still generalises. The trade-off did not die; it moved from the *class* to the *solution*.
- Naming what lives on the **complexity** side, and whether SGD minimises it, is unfinished. Hence *theories*, plural.

Next:

- Pick a paper, parse it using the lens of **fit-complexity**, if useful.
- Do a critical reading of each of the papers.
- Companion paper can either dispute these papers or validate them.



Backup: the Bayesian-evidence **complexity** term

Evidence (Bayesian model score, want it large): $p(S) = \int p(S | \mathbf{w}) p(\mathbf{w}) d\mathbf{w}$. The integrand peaks at $\hat{\mathbf{w}}$;
Laplace = peak height $\times$ width:

$$p(S) \approx \underbrace{p(S | \hat{\mathbf{w}}) p(\hat{\mathbf{w}})}_{\text{height}} \times \underbrace{(2\pi)^{p/2} (\det H)^{-1/2}}_{\text{width}} \implies -\ln p(S) \approx \underbrace{-\ln p(S | \hat{\mathbf{w}})}_{\text{fit}} + \underbrace{\frac{1}{2} \ln \det H - \ln p(\hat{\mathbf{w}})}_{\text{complexity}}.$$

$H = \nabla_{\mathbf{w}}^2 \left[-\ln p(S | \mathbf{w}) - \ln p(\mathbf{w}) \right] \Big|_{\hat{\mathbf{w}}}$ — curvature of the objective at the fit, $p \times p$, eigenvalues λ_k .

- Along eigen-direction k the fit stays good for $|\Delta w_k| \lesssim \lambda_k^{-1/2}$, so $\prod_k \lambda_k^{-1/2} = (\det H)^{-1/2} =$ **volume of weights that fit S** .
- Hence $\frac{1}{2} \ln \det H = -\ln(\text{that volume})$: *sharp* minimum $\rightarrow$ penalised, *flat* $\rightarrow$ rewarded — a soft *effective* parameter count, not the raw p .
- With $-\ln p(\hat{\mathbf{w}})$ (prior improbability of $\hat{\mathbf{w}}$):

$$\frac{1}{2} \ln \det H - \ln p(\hat{\mathbf{w}}) = -\ln \frac{p(\hat{\mathbf{w}})}{\sqrt{\det H}} \approx -\ln \frac{\text{good-fit volume}}{\text{prior volume}} = -\ln(\text{Occam factor}).$$
- Large n : $H \approx n\bar{H} \Rightarrow \frac{1}{2} \ln \det H \rightarrow \frac{p}{2} \ln n$ (Bayesian information criterion, BIC). In bits = $L(\mathbf{w})$ (MDL model bits). The *pair* is reparameterisation-invariant.

